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RESEARCH PAPER

Effects of tuning of ankle foot orthoses-footwear combination using wedges on stance phase knee hyperextension in children with cerebral palsy – Preliminary results

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Abstract

Purpose. This pilot study investigated the feasibility of reducing stance phase knee hyperextension in children with cerebral palsy by tuning the ankle foot orthoses-footwear combination (AFO-FC) using different sizes of wedges.

Methods. Five children with cerebral palsy underwent three dimensional gait analysis and tuning of their AFO-FC using wedges. Data analysis was carried out by comparing relevant gait parameters between the non-tuned and tuned prescription.

Results. Knee hyperextension during stance significantly decreased, and the shank to vertical angle was closer to normal after tuning. Although none of the other parameters showed statistically significant changes, the wide confidence intervals and lack of power indicated the likelihood of a type II error. Further, it was noted that the influence of tuning on temporal-spatial parameters was different between children with diplegia and those with hemiplegia. It was estimated that a sample size of 15 is required to detect significant changes at $p = 0.05$ and power of 0.8.

Conclusions. The findings of this study clearly indicate the potential clinical utility of tuning using wedges to correct knee hyperextension during the stance phase in children with cerebral palsy. However, observations support the need for an adequately powered study to assess the long-term effects of tuning on gait parameters, activity level and quality of life.

Keywords: *Tuning, cerebral palsy, ankle foot orthoses, gait, gait analysis*

Introduction

Cerebral palsy predominantly affects movement and posture and hence influences ambulation. Although approximately two-thirds of all children with cerebral palsy achieve some degree of ambulation, their walking patterns typically vary from those with non-impaired gait [1]. Rigid ankle foot orthoses (AFOs) are among the most commonly prescribed devices to manage gait impairments of ambulatory children with cerebral palsy. However, ambiguity exists in the literature regarding the effectiveness of AFOs for children with cerebral palsy. While one study found no significant changes except at the ankle joint [2], others noted statistically significant changes in temporal and spatial parameters [3–5] as well as changes in kinematics of proximal joints [4,6]. Such ambiguity may be attributed to several factors. First, there is lack of uniformity in study designs and sample characteristics [7], second, the lack of comparisons that are specific to different diagnostic groups. The only two studies that compared the effects of AFO intervention on children with hemiplegia and diplegia [5,8] produced conflicting results. Finally, none of the studies considered biomechanical optimisation of AFOs, which is considered a critical influence in AFO intervention [9].

The potential of biomechanical optimisation of AFOs, or "tuning" [9], was identified more than 20 years ago [10,11]. However, there are only a few reports on the role of tuning in AFO intervention.

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Tuning involves modifying the alignment of AFO – footwear combination (AFO-FC) to optimise kinetics and kinematics of gait. The parameters that are commonly modified include height of the heel, type and design of the heel and type of the rocker at the metatarsal heads [9,12,13]. These are hypothesised to influence initial stance, mid-stance and terminal stance, respectively [12]. Although a few studies have looked at the effects of tuning as a whole on gait parameters of children with cerebral palsy [14,15], no studies have considered the influence of individual components of tuning.

Although Stallard and Woollam [14] hypothesised that tuning influences the knee joint by controlling ankle joint kinematics, Butler et al. [15] found that appropriately tuned, rigid AFOs directly modify the orientation of the ground reaction force (GRF) in relation to the proximal joints. They claimed that this would reduce the abnormal moments acting on the knee and hip, thereby decrease knee hyperextension and improve hip flexion. They also stated that properly aligned AFOs prevent ankle joint motion, passively maintaining the forces acting on the knee joint, and thereby optimising knee kinematics. This relieves the child from the effort of maintaining knee alignment, thus providing an environment to help motor learning to occur. Condie and Meadows [16] made two interesting suggestions: first, that tuning controls the forward progression of centre of pressure, thereby facilitating push-off. Second, they hypothesised that inappropriate AFO-FC can lead to increased energy expenditure in comparison to barefoot gait, although neither idea was investigated.

One of the common gait abnormalities seen in children with cerebral palsy is hyperextension of the knee joint during stance phase, resulting from a lack of progression of the tibia over the stable foot during stance phase [17]. An appropriate heel raise (wedge) may modify the shank to vertical angle (SVA), which is the angle formed by the shank of the tibia, to an imaginary line perpendicular to the ground. This, in turn, may modify the orientation of the GRF in relation to the proximal joints. If done judiciously, this can optimise mid-stance shank kinematics, and thereby gait [12]. The SVA and the orientation of the GRF have been suggested as key indicators for tuning [12,18].

Butler et al. [15] investigated the effect of tuned AFOs in conjunction with balance training exercises. Their study included five children with cerebral palsy who had hyperextension of the knee joint and an increased knee extending moment arm during mid stance. Gait assessment and clinical examination were conducted before treatment and between 4 and 6 months later. The results showed that the high knee extending moment arm was significantly decreased ($p < 0.01$) and was closer to normal. There was no significant change in walking speed. Three of five children retained the improvement in barefoot walking and were weaned from their AFOs. Although the study reported some effects of tuning, joint kinematics were not considered. The authors did not explain the procedure of tuning and the data were collected while the children walked barefoot, rather than while wearing orthoses.

To summarise, ambiguity exists in the literature regarding the effectiveness of rigid AFOs in children with cerebral palsy. This may be attributed to differences in sample characteristics and study designs. There is also lack of evidence relating to tuning of AFO-FCs for children with cerebral palsy. No studies have yet investigated the use of any specific component of tuning on a particular gait abnormality, such as hyperextension of the knee during stance phase. In this pilot study, the first aim was to investigate the feasibility of decreasing stance phase hyperextension by tuning using different sizes of wedges. The second aim was to determine the sample size to detect a clinically significant difference for an adequately powered future study investigating the effects of tuning on the gait of children with cerebral palsy.

Methods

Subjects

Five children with cerebral palsy (three females and two males) were included in the study, two of whom were diagnosed as having spastic diplegia, and three of whom had spastic hemiplegia. The mean age (SD) of the participants was 9.7 (3.5) years; detailed sample characteristics are provided in Table I. Ethical approval was gained from the local UK NHS research ethics committee and informed consent was obtained from the participants and their parents. All the children demonstrated hyperextension of the knee ($< 0^\circ$ of flexion) during the stance phase of gait. The children with hemiplegia were using rigid AFOs on their affected legs, and one child with diplegia was using a rigid AFO on the right leg and a dynamic AFO on the left leg. While the other child with diplegia was using rigid AFOs on both

Table I. Sample characteristics.

	Age	Sex	Diagnosis	Leg using rigid AFO
Subject 1	5.6	Male	Diplegia	Right
Subject 2	12.5	Female	Hemiplegia	Right
Subject 3	6.2	Female	Hemiplegia	Right
Subject 4	12.6	Female	Diplegia	Right
Subject 5	11.8	Male	Hemiplegia	Left

legs, only the right leg was considered, because the left knee did not demonstrate hyperextension during stance. None of the subjects had received botulinum toxin injections within a 6-month period before starting the study and none had undergone surgery.

Equipment

Gait assessment was conducted using a three-dimensional Vicon 612 motion analysis system with eight cameras (Vicon motion systems, Oxford, UK), and two force plates (AMTI, Watertown, MA) embedded in the middle of the walkway in the gait analysis laboratory. The Vicon Workstation software was used as the control interface for capture, reconstruction, editing and processing of the data. Twenty-two reflective markers were placed on the anatomical landmarks of the participants' trunk, pelvis and lower limbs as required by the Plug in Gait model provided by Vicon motion systems. The wedges used for tuning were made of non-deformable ethyl vinyl acetate (EVA). They were custom-made by the orthotist according to the shoe size of the child being tested. The wedges were categorised based on the angle made by the top surface of the wedge to the perpendicular. The height, length and inclination of the wedge were estimated to produce the required angle for each shoe size.

Procedure

Gait analysis was carried out with children walking at a self-determined speed and wearing their own AFO-FC (non-tuned). This was followed by tuning using wedges, and analysis of their gait wearing the final tuned prescription. The wedges were attached to the soles of the shoes using double sided tape and elastic adhesive bandages. The placements of wedges were as close to the heel as possible and in line with the long axis of the shoes (Figure 1). The procedure was carried out in a single visit with only the time required to attach the wedges (less than 5 min) as recovery time between the conditions. No consistent recovery time was given between the trials, although adequate rest periods were given according to the child's requirement. The protocol aimed to capture three clean strikes on the force plates by the leg using the rigid AFO. Walks were repeated until this was achieved. The SVA was measured in standing using a manual goniometer while the child was wearing original AFO-FC prescription (Figure 1). Wedges were attached to the heels of the shoes until the SVA during standing was approximately 12° inclined, which is considered normal [12]. Taking this as a starting point, data were collected and the SVA was further adjusted by adding or removing wedges until the GRF in the middle of mid-stance was in optimal alignment in relation to the knee, i.e., as close as possible to the centre of the knee joint (Figure 2). The same sizes of wedges were added to the contra-lateral limb to maintain symmetry. Once the tuning was finalised, data were collected for the final prescription.

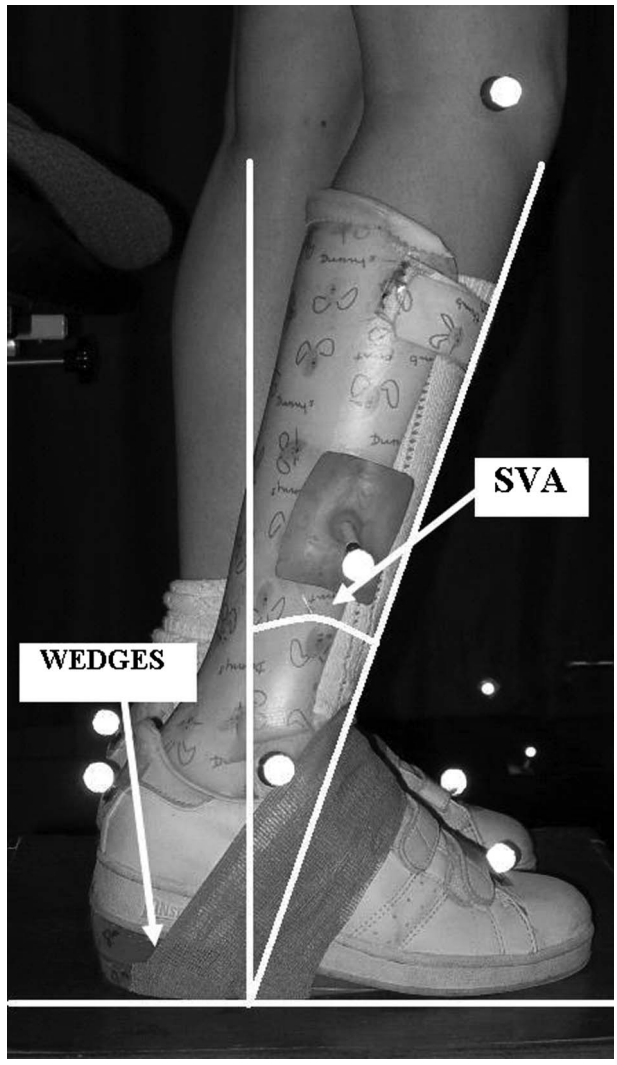

Figure 1. Wedge attached to the shoes using double sided tape and elastic bandages and SVA.

Analysis was carried out by comparing data from the tuned leg between the original (non-tuned) and final (tuned) prescriptions, including:

- relevant temporal and spatial parameters,
- kinematic variables such as knee flexion at initial contact, and peak knee flexion and peak knee extension during stance and
- kinetic variables such as peak knee flexion moments and peak knee extension moments during stance.

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Statistical comparisons of the data were conducted using paired *t*-test or Wilcoxon signed rank test, depending on the distribution of the data. Sample size is one of the factors which influence calculation of p [19], hence testing for statistical significance is not considered ideal for small samples. However, as this is a pilot study, it was considered important to explore the statistical significance of the comparisons conducted. Significance testing was supplemented with 95% confidence intervals and power analysis to evaluate the possibility of type II error [20,21]. In addition, effect sizes [22] were used to avoid making erroneous inferences (type I error) based on significant differences [19]. This combination of analysis techniques provided valuable information. Owing to the small sample size the 95% confidence intervals was determined based on the *t*-distribution [21]. The sample size required to detect a medium effect size for peak knee flexion during stance at $p = 0.05$ and power of 0.8 was also determined [21].

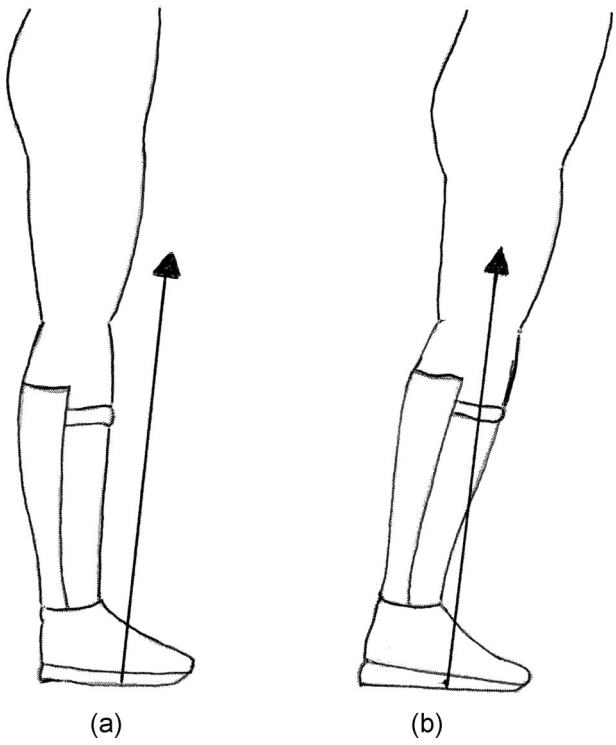

Figure 2. Optimisation of GRF: (a) before tuning and (b) after tuning.

Statistical analyses were carried out using SPSS for Windows version 15.0.1.

Results

Temporal and spatial parameters

The means (SD) of the temporal-spatial parameters for tuned and non-tuned AFO-FCs are given in Table II with statistical comparisons. None of the differences were significant. When compared separately, children with diplegia had lower mean values of 15% for velocity, 9% for cadence and 6% for stride length for the tuned condition compared to non-tuned. However, children with hemiplegia showed 7% higher mean values in the tuned condition compared to non-tuned for velocity and cadence, while the stride length remained the same (Table III).

Shank to vertical angles

The mean (SD) of SVA measured in standing with non-tuned and tuned AFO-FCs are given in Table IV. The mean SVA was about 5° higher with tuned AFO-FCs in comparison to the non-tuned, which represented a statistically significant difference ($p = 0.005$).

Knee kinematics

The knee joint angles at specific points during the stance phase were compared between tuned and non-tuned AFO-FC. The variables considered are given in Table V. The results showed a tendency towards greater flexion throughout the stance phase with tuned AFO-FC, compared to non-tuned (Figure 3). Although the knee flexion at initial contact and maximum knee flexion during stance phase were higher in the tuned condition (17.2 and 25.2, respectively) compared to non-tuned (13.7 and 19.7, respectively), the changes were not statistically significant. However, the mean maximum knee extension during stance phase was significantly ($p = 0.04$) lower, at 3.7° of flexion when tuned,

Table II. Temporal-spatial parameters with tuned and non-tuned AFO-FC.

Parameters	Non tuned AFO-FC (mean (SD))	Tuned AFO-FC (mean (SD))	95% Confidence interval of the difference	Test	*p* value
Velocity (m/s)	1.08 (0.1)	1.07 (0.14)	−0.17 to 0.2	Paired *t*-test	0.80
Cadence (steps/min)	122.5 (16.6)	122.3 (12.4)	−16.3 to 16.8	Paired *t*-test	0.97
Stride length (m)	1.08 (0.19)	1.06 (0.20)	−0.06 to 0.1	Paired *t*-test	0.54

Table III. Mean (SD) of temporal-spatial parameters based on the types of cerebral palsy for non-tuned and tuned AFO-FCs.

Types of CP	Conditions	Parameters		
		Velocity (m/s)	Cadence (steps/min)	Stride length (m)
Diplegia	Non tuned AFO-FC (mean (SD))	1.10 (0.18)	136.9 (18.1)	0.98 (0.29)
	Tuned AFO-FC (mean (SD))	0.94 (0.09)	124.3 (17.7)	0.92 (0.22)
Hemiplegia	Non-tuned AFO-FC (mean (SD))	1.08 (0.14)	113.0 (6.7)	1.15 (0.12)
	Tuned AFO-FC (mean (SD))	1.15 (0.07)	121.0 (12.0)	1.15 (0.14)

Table IV. SVA with non-tuned and tuned AFO-FCs.

	Non-tuned AFO-FC (mean (SD))	Tuned AFO-FC (mean (SD))	95% Confidence interval of the difference	Test	p value
SVA (degrees)	5.6 (3)	10.8 (1.8)	−7.7 to −2.6	Paired t-test	0.005*

*Significant difference between the two conditions at $p < 0.05$.

Table V. Kinematic parameters of the knee joint in degrees with non-tuned and tuned AFO-FCs.

Parameters	Non tuned AFO-FC (mean (SD))	Tuned AFO-FC (mean (SD))	95% Confidence interval of the difference	Test	p value
Knee flexion (initial contact)	13.7 (8.4)	17.2 (5.1)	−11.5 to 4.4	Wilcoxon signed ranks	0.14
Max. knee flexion (stance)	19.7 (9.3)	25.2 (5.3)	−11.7 to 0.6	Paired t-test	0.06
Max. knee extension (stance)	−2.6 (2.8)	3.7 (3.3)	−10.2 to −2.5	Wilcoxon signed ranks	0.04*

*Significant difference between the two conditions at $p < 0.05$.

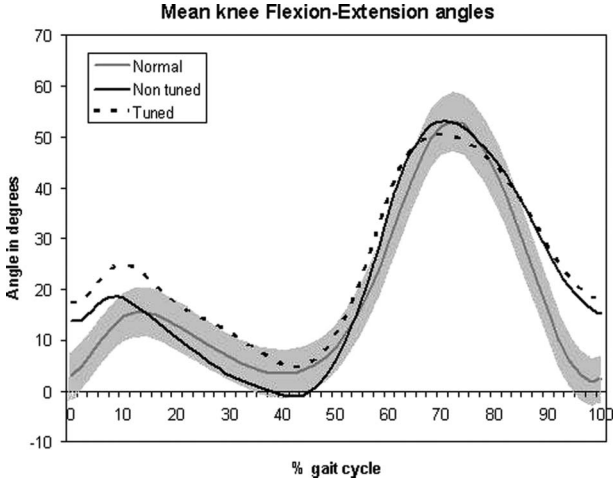

Figure 3. Graph comparing mean flexion-extension of the knee joint between tuned and non-tuned AFO-FCs during one complete gait cycle. The shaded area denotes one standard deviation from the normal.

compared with 2.6° of hyperextension in the non-tuned condition.

Kinetic parameters

The kinetic parameters showed several non-significant differences between the tuned and non-tuned conditions (Table VI). The mean peak knee flexion moments tended to be higher in the tuned condition (0.7 Nm/kg), compared to non-tuned (0.59 Nm/kg) and peak knee extension moments tended to be less extending in the tuned condition (−0.29 Nm/kg) compared to non-tuned (−0.44). These differences were not statistically significant (Table VI). Similar to the knee kinematics, the sagittal plane knee moments also showed a tendency towards increased flexion throughout the stance phase with tuned AFO-FC, compared to non-tuned.

Power analysis

Standardised effect size [22] and power were calculated for all the non-significant parameters to determine the probability of making a type II error (Table VII). Temporal and spatial parameters, knee flexion at initial contact and the peak knee flexion moment had small effect sizes. In comparison, high effect sizes were demonstrated by the peak knee extension during stance, maximum knee flexion during stance and SVA. Finally, the peak knee extension moment during stance phase had a medium effect size. The results of peak knee extension during stance and SVA (parameters with

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Table VI. Sagittal plane moments (Nm/kg) acting on the knee joint with tuned and non-tuned AFO-FCs.

Parameters	Non-tuned AFO-FC	Tuned AFO-FC	95% Confidence interval of the difference	Test	p value
Peak knee flexion moment (stance)	0.59 (0.31)	0.7 (0.32)	−0.35 to 0.11	Paired t-test	0.25
Peak knee extension moment (stance)	−0.44 (0.2)	−0.29 (0.24)	−0.39 to 0.09	Wilcoxon signed ranks	0.14

Table VII. Standardised effect size and power of the knee kinematics, kinetics and SVA.

Variable	Effect size	Power
Velocity	−0.09	0.06
Cadence	−0.01	0.05
Stride length	−0.24	0.08
Knee flexion (initial contact)	0.39	0.16
Max. knee flexion (stance)	0.80	0.50
Peak knee extension (stance)	1.44	0.92
Peak knee flexion moment (stance)	0.43	0.19
Peak knee extension moment (stance)	0.56	0.28
SVA	1.79	0.93

significant differences between the tuned and non-tuned conditions) had powers ($1 - \beta$) greater than 0.9. Among the parameters with non-significant differences between the tuned and non-tuned conditions, the results of maximum knee flexion had a power of 0.5, whereas all the other variables had a power of less than 0.3 (Table VII). At $p < 0.05$ and a power of 0.8, it was determined that a sample of 15 will be adequate to detect a change of 3.5 degrees of peak knee flexion during stance.

Discussion

No significant differences in the temporal and spatial parameters were evident when comparing tuned and non-tuned conditions. However, Table III demonstrates that while the temporal-spatial parameters of children with hemiplegia showed a tendency to improve with tuning, that of children with diplegia demonstrated a tendency to deteriorate with tuning. This tendency to deteriorate may reflect a lack of ability in children with diplegia to get accustomed to the new prescription in which their usual alignment was disturbed compared to children with hemiplegia. However, there is no evidence in the literature to support a difference in ability.

The results of this study might have been affected by the fact that no time was given to the participants to get familiarised with the new prescription. Previous studies that investigated the effects of AFOs on the gait of children with cerebral palsy [5,8,22] provided the participants with 4–6 weeks for familiarisation with the prescription. The results may have been different after a period of familiarisation with the new prescription. Such a possibility is supported by findings from a case study involving an adult with hemiplegia in which the effects of tuning were investigated over a period of 3 months [23]. In this case, it was reported that while there was no improvement in temporal-spatial parameters immediately after tuning, there was increase in velocity, step length and single support time after 3 months.

Another explanation for the deterioration in temporal-spatial parameters in children with diplegia may be that tuning was not as effective for children with diplegia compared to children with hemiplegia. However, the sample size of children with diplegia in this study was too small, and no study has investigated effects of tuning on children with diplegia alone. In this study, no conclusions can be made without a long term study and with a bigger sample. Nevertheless, the tendencies seen in this study and existing ambiguity in the literature on AFO intervention in cerebral palsy [5,8] indicate the importance of group-wise comparisons based on diagnosis when investigating the effects of interventions like AFO and tuning of AFO-FC in children with cerebral palsy.

The SVA significantly increased after tuning compared to non-tuned AFO-FC (Table IV). The change in SVA had an effect size of 1.8 (Table VII), which also supports the result from significance testing. In this study, the SVA was only used as an indicator of the more appropriate size of the initial wedge to start the tuning process. The final mean SVA of the tuned AFO-FCs was 10.8, which corroborates with the findings of Owen [24] who reported a mean (SD) SVA of 11.86 (2.05) for tuned AFO-FCs in 69 children with cerebral palsy.

When comparing three kinematic variables in this study, knee hyperextension during the stance phase was significantly less for the tuned condition compared to non-tuned ($p = 0.04$). The large effect size (1.44) of the change in peak knee extension supported this finding (Table VII). The significantly lower knee hyperextension suggests the effectiveness of wedges in controlling hyperextension of the knee joint during stance phase in children with cerebral palsy. This can be attributed to the optimisation of GRF and shank kinematics, in line with the suggestions of Butler and Nene [18] regarding the use of

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heel raises to reorient the GRF closer to the knee joint centre in children with cerebral palsy. In this study, all the children had GRF vector oriented anterior to the knee joint during mid-stance with their original prescription which then produced an extension moment and thereby hyperextension at the knee. Optimisation of GRF vector reduced the extension moment at the knee for all the children which probably resulted in reduced knee hyperextension. However, the difference in peak knee extension moment at the knee was not statistically significant, the wide 95% confidence interval for the difference between tuned and non-tuned AFO-FCs (-0.39 to 0.09) indicates the need for a study with a larger sample size. It has also been reported that tuning using wedges can produce unwanted initial stance flexion [25]. In this study, the increase in knee flexion of 3.6° at initial contact and 5.6° in stance from non-tuned to tuned conditions were not statistically significant. However, these differences may be clinically relevant. Furthermore, the wide 95% confidence interval suggests the possibility of lack of power in the sample [20].

Despite the obvious limitations in sample size that is characteristic of most studies involving children with cerebral palsy, this study offers useful information regarding research design considerations such as effect sizes, power and sample sizes. On the basis of Cohen's classification [22], three of the parameters had large effect sizes, one medium and five low. The temporal and spatial parameters had negative effect sizes because they showed a tendency to be lower in the tuned than non-tuned conditions. The lack of power (<0.8) and wide 95% confidence interval suggested that all non-significant variables were prone to type II error. Sample size calculation was based on peak knee flexion during stance, as it yielded the highest effect size among the non-significant parameters. It was estimated that a change of 3.5° would be adequate to yield a medium effect size. Therefore, a future study with a sample size of 15 will be powerful enough to detect significant differences at $p = 0.05$ and power of 0.8.

Although the results of this study showed a significant improvement in knee hyperextension with tuning, it is not yet clear how this difference might influence the overall gait pattern of children with cerebral palsy over time. The lack of improvement in temporal and spatial parameters also warrants a long-term study. The effect of increased flexion on initial stance is also unclear. Although Butler et al. [25] suggested that the overall benefits of tuning offset the disadvantage of increased initial stance flexion, the picture is incomplete without further investigation. It is necessary to study the effects of changes on the overall gait pattern, muscle characteristics such as flexion contractures, activity and quality of life of children over time. It is also important to explore the impacts of other components of tuning, such as heels for initial stance, or rockers for terminal stance, as well as considering variables relating to other lower limb joints and the trunk.

Conclusion

The findings of this pilot study clearly indicate the potential clinical utility of tuning using wedges to correct knee hyperextension during stance phase in children with cerebral palsy. Although the use of wedges was found to be effective for hyperextension of the knee joint during stance phase, tuning remains incomplete without consideration of other stages of the stance phase and other joints. Observations support the need for an adequately powered study to be conducted to assess the effects of tuning on knee moments and flexion during initial stance. In addition, the GRF and long-term effects of tuning on gait parameters, activity level and quality of life need to be investigated

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